

December 18, 2025

1. By how much does the emitted current change if the tungsten cathode temperature is increased from  $T = 2250$  K to  $2400$  K? The work function for tungsten is  $W_{\text{ki}} = 5.0$  eV. The Richardson coefficient is  $6.05 \cdot 10^5$  A/m<sup>2</sup>K<sup>2</sup>.

The emission current density is given by the Richardson–Dushman equation

$$j = AT^2 \exp\left(-\frac{W_{\text{ki}}}{kT}\right).$$

The ratio of the emitted current densities is

$$\frac{j_2}{j_1} = \frac{T_2^2}{T_1^2} \exp\left[-\frac{W_{\text{ki}}}{k} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)\right].$$

Substituting the numerical values,

$$\frac{j_2}{j_1} = \frac{2400^2}{2250^2} \exp\left[-\frac{5.0 \cdot 1.6 \cdot 10^{-19}}{1.38 \cdot 10^{-23}} \left(\frac{1}{2400} - \frac{1}{2250}\right)\right].$$

The emitted current increases by approximately a factor of

$$\frac{j_2}{j_1} \approx 8.$$

2. What retarding voltage reduces the emission current density to one fifteenth for a cathode at  $T = 2300$  K? The electron charge is  $q = -1.6 \cdot 10^{-19}$  As.

The current density in the presence of a retarding voltage  $U$  is

$$j = j_0 \exp\left(\frac{qU}{kT}\right).$$

The condition

$$\frac{j}{j_0} = \frac{1}{15}$$

leads to

$$\exp\left(\frac{qU}{kT}\right) = \frac{1}{15}.$$

Taking the natural logarithm,

$$\frac{qU}{kT} = \ln\left(\frac{1}{15}\right).$$

Solving for the retarding voltage,

$$U = \frac{kT}{q} \ln\left(\frac{1}{15}\right).$$

Substitution yields

$$U = \frac{1.38 \cdot 10^{-23} \cdot 2300}{1.6 \cdot 10^{-19}} \ln\left(\frac{1}{15}\right),$$

which gives

$$U \approx -0.55 \text{ V}.$$

3. What is the free-electron drift velocity in a copper conductor of radius  $R = 2.5 \text{ mm}$  if the current is  $I = 15 \text{ A}$ ? The electron density in copper is  $n_e = 3.3 \cdot 10^{28} \text{ m}^{-3}$ .

The current is related to the drift velocity by

$$I = n_e q v_d A,$$

where the cross-sectional area is

$$A = \pi R^2.$$

Solving for the drift velocity,

$$v_d = \frac{I}{n_e q \pi R^2}.$$

Substituting the given values,

$$v_d = \frac{15}{3.3 \cdot 10^{28} \cdot 1.6 \cdot 10^{-19} \cdot \pi \cdot (2.5 \cdot 10^{-3})^2}.$$

The drift velocity is

$$v_d \approx 1.4 \cdot 10^{-4} \text{ m/s}.$$

4. We examine an n-type semiconductor under the conditions shown in the figure:  $B_y = 0.5 \text{ T}$ ,  $U_H = 12 \text{ mV}$ ,  $I = 25 \text{ mA}$ . The magnetic induction is directed along the  $y$ -axis.

- (a) What is the electron concentration?

The Hall voltage is given by

$$U_H = \frac{I B_y}{n q d},$$

where  $d$  is the sample thickness.

Solving for the electron concentration,

$$n = \frac{I B_y}{q U_H d}.$$

- (b) What is the drift velocity of the electrons?

The drift velocity follows from the Hall field:

$$v_d = \frac{U_H}{B_y d}.$$

- (c) What is the Hall constant?

The Hall constant is defined as

$$R_H = \frac{1}{n q}.$$